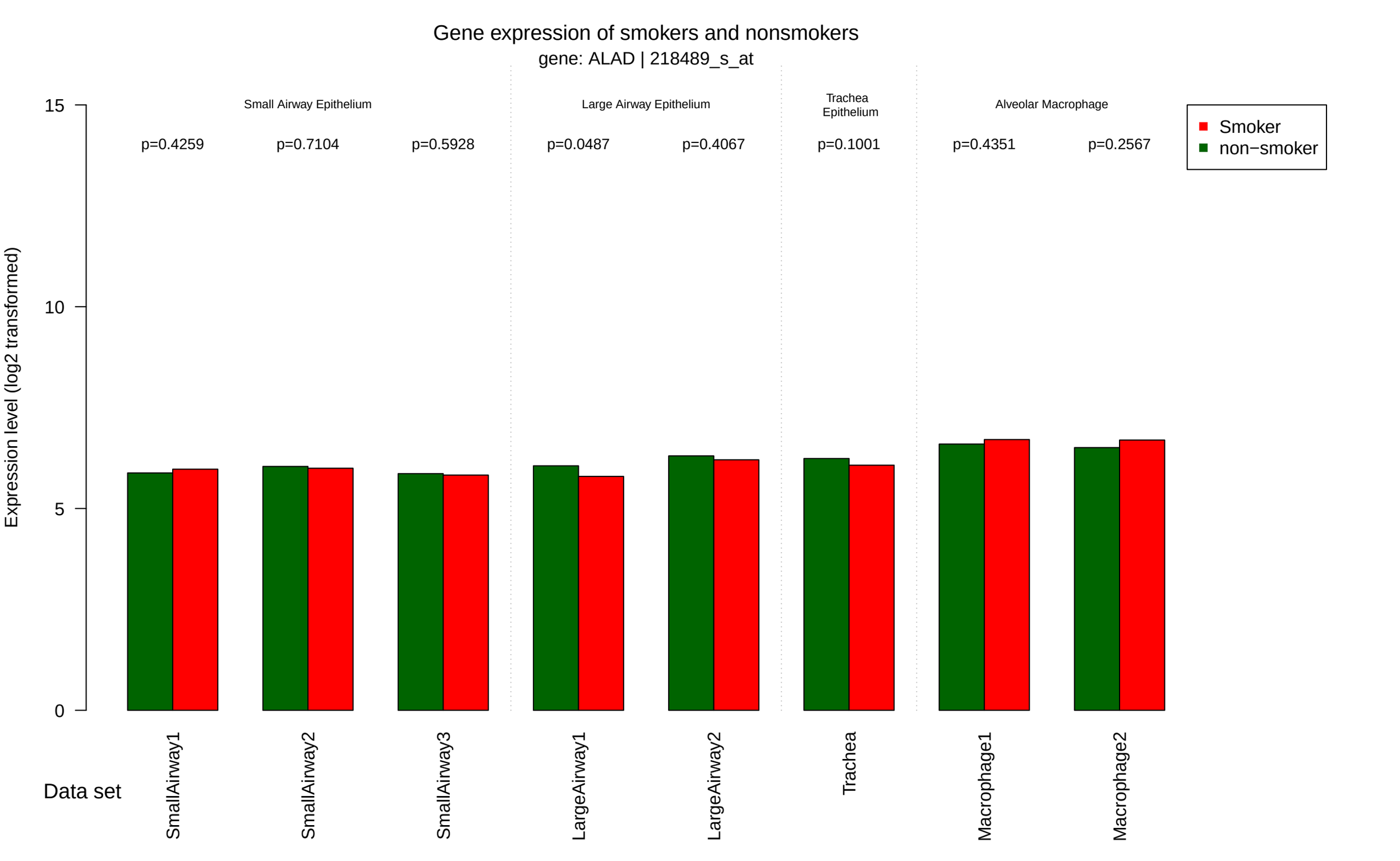
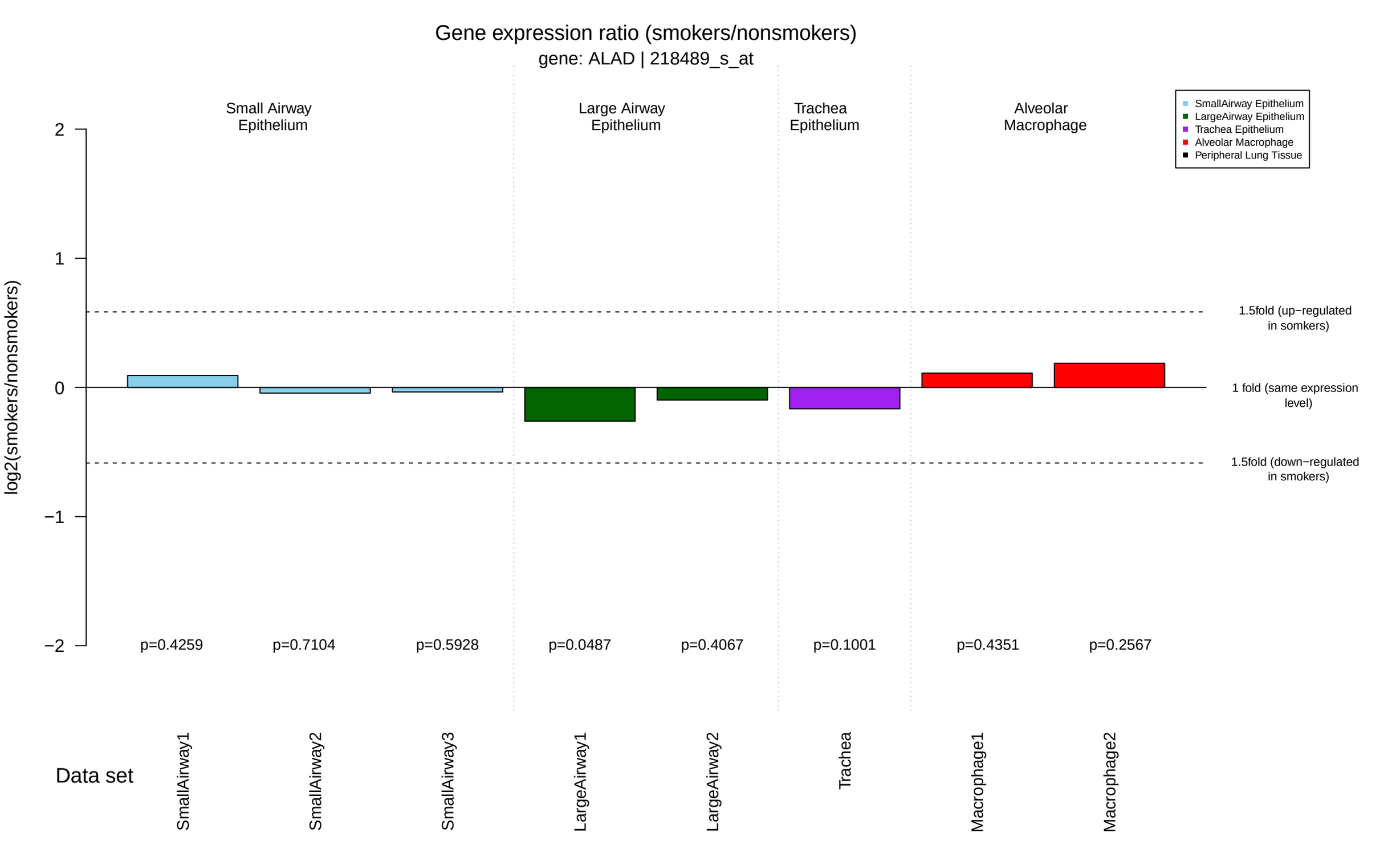
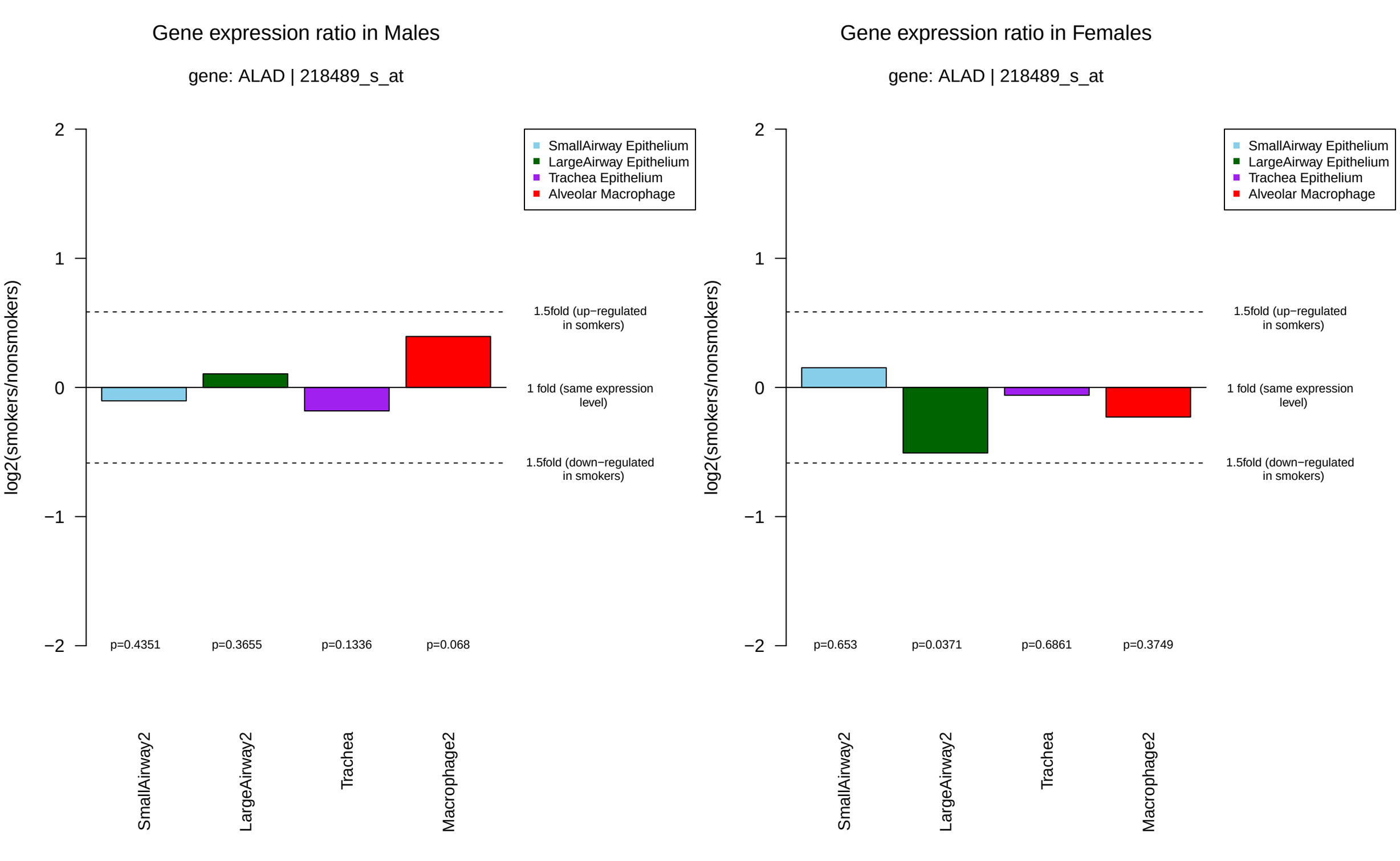
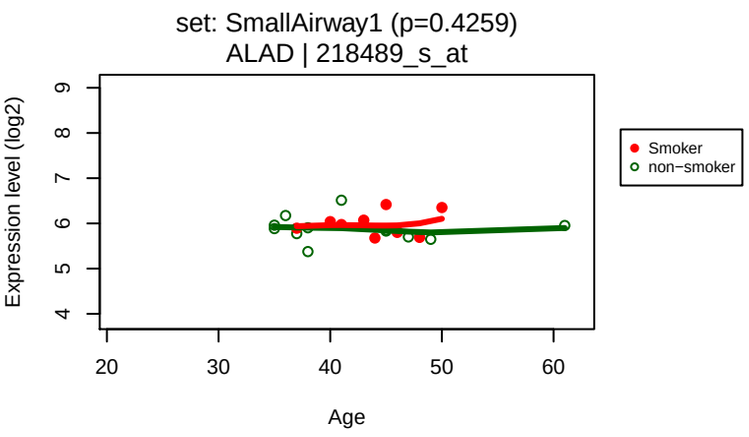
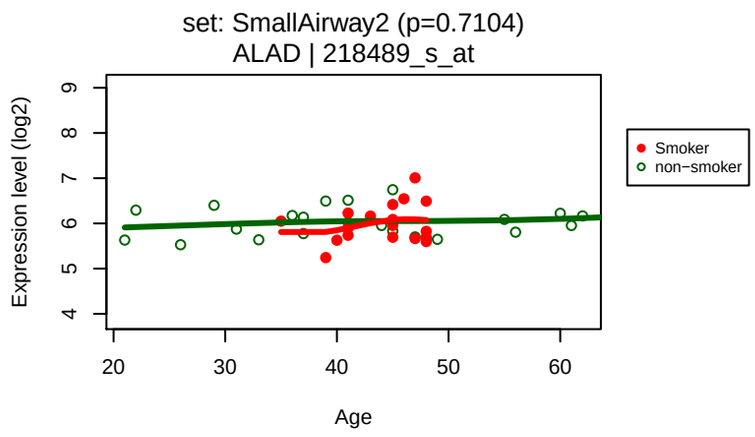
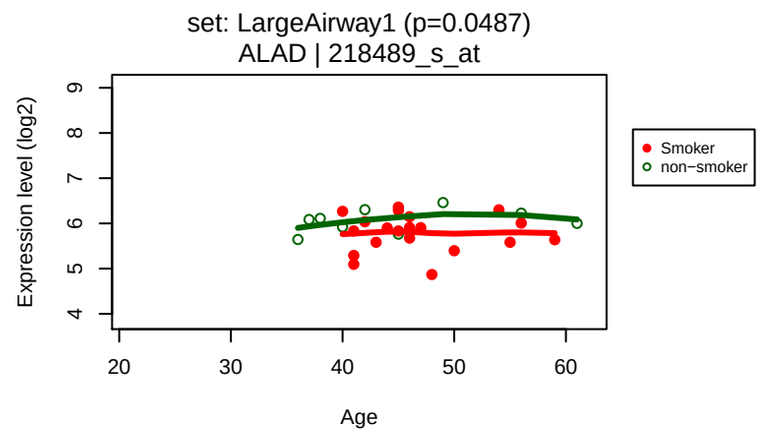

Gene expression of smokers and nonsmokers
gene: ALAD | 218489_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4259
p=0.7104
p=0.5928
p=0.0487
p=0.4067
p=0.1001
p=0.4351
p=0.2567
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

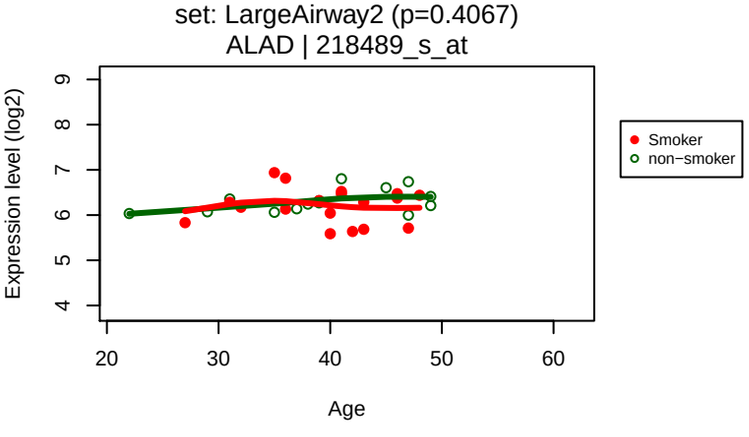

Gene expression ratio (smokers/nonsmokers)
gene: ALAD | 218489_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4259
p=0.7104
p=0.5928
p=0.0487
p=0.4067
p=0.1001
p=0.4351
p=0.2567
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

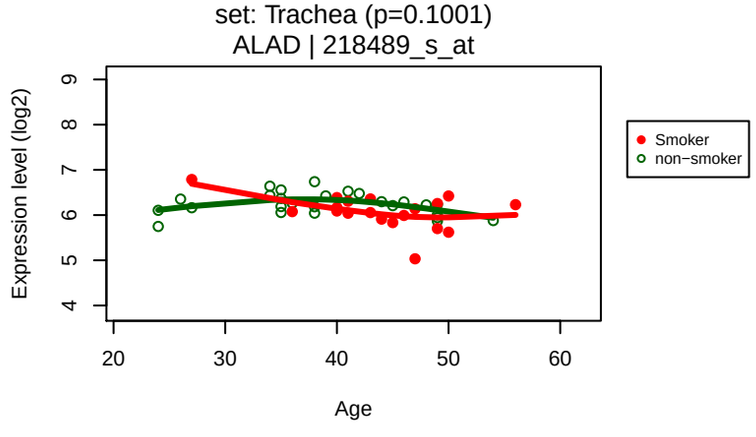

Gene expression ratio in Males
gene: ALAD | 218489_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4351
p=0.3655
p=0.1336
p=0.068
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALAD | 218489_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.653
p=0.0371
p=0.6861
p=0.3749
SmallAirway2
LargeAirway2
Trachea
Macrophage2

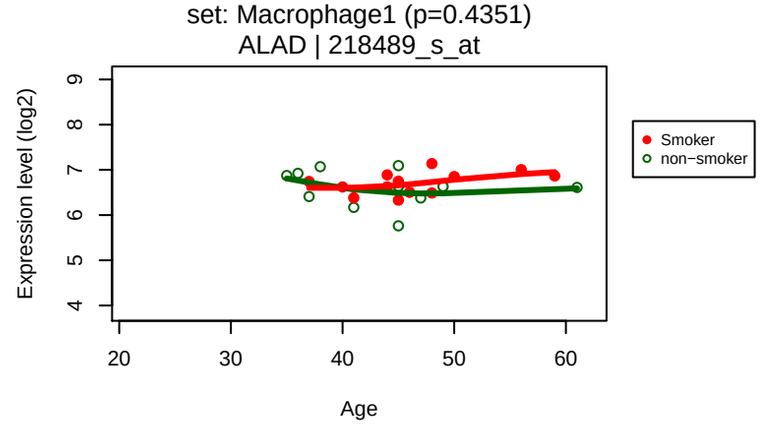

set: SmallAirway1 (p=0.4259)
ALAD | 218489_s_at
Expression level (log2)
Age
Smoker
non−smoker

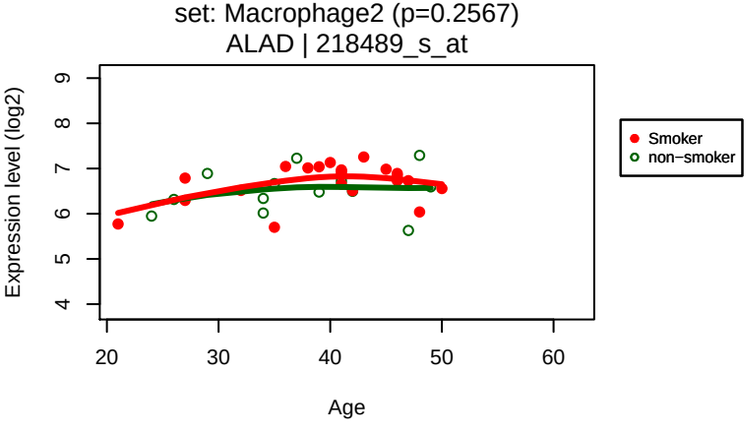

set: SmallAirway2 (p=0.7104)
ALAD | 218489_s_at
Expression level (log2)
Age
Smoker
non−smoker

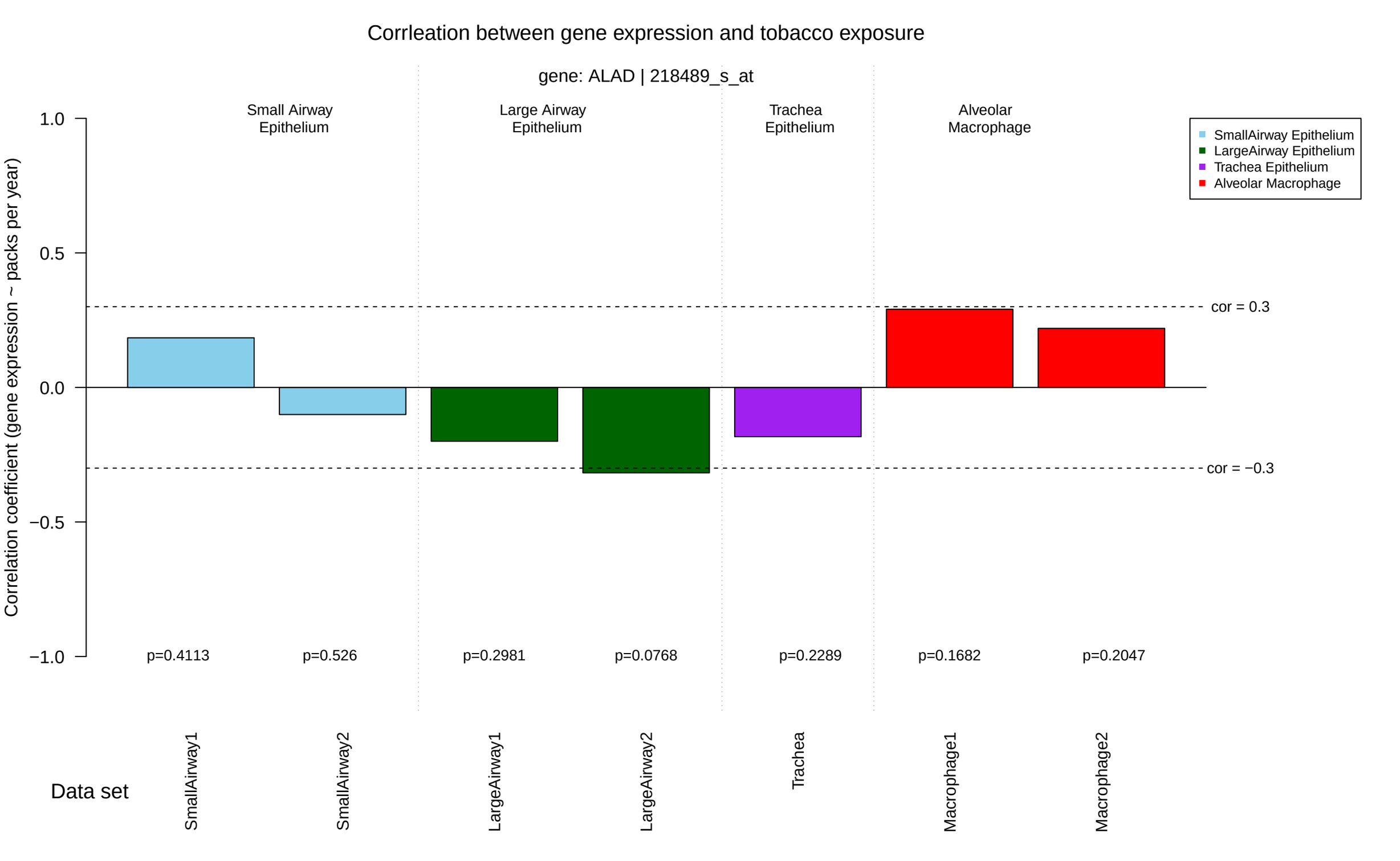

set: LargeAirway1 (p=0.0487)
ALAD | 218489_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4067)
ALAD | 218489_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1001)
ALAD | 218489_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4351)
ALAD | 218489_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2567)
ALAD | 218489_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALAD | 218489_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4113
p=0.526
p=0.2981
p=0.0768
p=0.2289
p=0.1682
p=0.2047
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2